

Stage 4 Report

Integrated Robot Motion Planning and Pick-and-Place Simulation System

1. Introduction

The objective of Stage 4 is to integrate the previously developed robot kinematics, trajectory planning, obstacle avoidance, and robot control modules into a complete industrial robot manipulation system.

In this stage, a full pick-and-place simulation system is developed using PyBullet and a Franka Panda 7-DOF robotic manipulator. The robot is capable of automatically moving to the object, grasping the object, generating a collision-free trajectory, avoiding obstacles, placing the object at the target position, and returning to the home position.

The final system demonstrates the integration of motion planning, inverse kinematics, obstacle avoidance, and task scheduling for an industrial robotic transportation task.

2. System Architecture

The integrated system consists of several functional modules, including task scheduling, motion planning, obstacle avoidance, inverse kinematics, robot control, and simulation environment.

The complete system architecture is shown in Figure 1.

System Architecture of Integrated Robot Pick-and-Place System

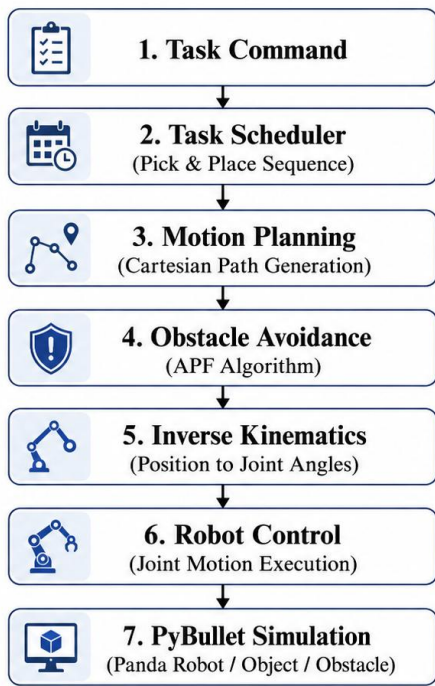
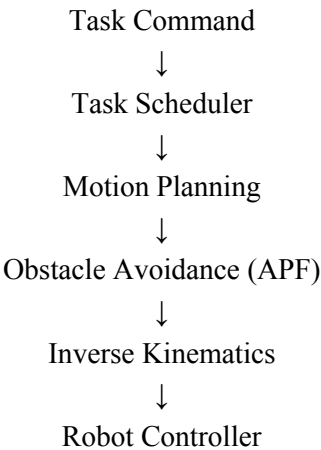


Figure 1. System architecture of the integrated robot pick-and-place simulation system.

The workflow of the system is:



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PyBullet Simulation Environment

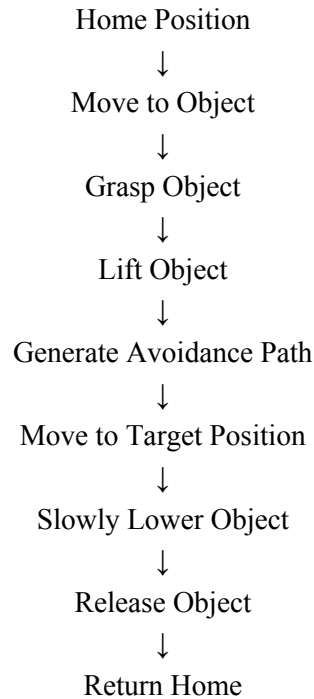
Each module communicates with the next module to complete the entire robotic manipulation process.

3. System Components

1) Task Scheduler Module

The task scheduler controls the overall operation sequence of the robot.

The designed industrial transportation workflow is:



The scheduler ensures that each action is executed in the correct order and provides automatic task execution without manual control.

2) Robot Simulation Module

The robot simulation module is implemented using PyBullet.

The simulation environment contains:

- Franka Panda 7-DOF robotic arm
- Working table
- Object to be transported
- Static obstacle
- Pick position
- Place position

The PyBullet environment provides realistic visualization and physics simulation for evaluating robot motion performance.

3) Inverse Kinematics Module

Inverse kinematics is used to convert desired Cartesian end-effector positions into robot joint angles.

The input of the module is:

End-effector Cartesian Position (x, y, z)

The output is:

Robot Joint Configuration ($q_1, q_2, q_3, q_4, q_5, q_6, q_7$)

The calculated joint positions are then sent to the robot controller for execution.

4) Trajectory Planning Module

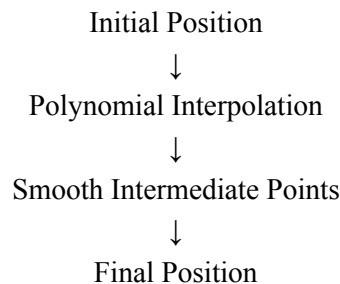
To achieve smooth robot movement, quintic polynomial interpolation is used for trajectory generation.

The quintic trajectory provides smooth:

- Position
- Velocity
- Acceleration

Compared with direct point-to-point motion, the generated trajectory reduces sudden movement changes and improves robot stability.

The trajectory planning process is:



5) Obstacle Avoidance Module

An Artificial Potential Field (APF) algorithm is implemented to achieve automatic obstacle avoidance.

The APF method consists of two forces:

Attractive Force

The target position generates an attractive force:

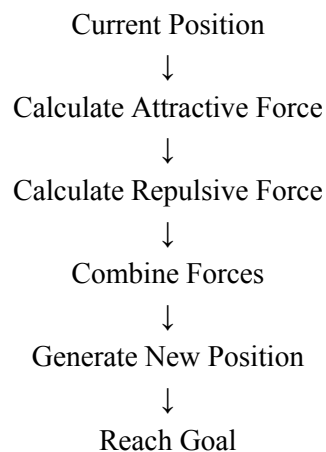
- Guides the robot toward the goal
- Ensures task completion

Repulsive Force

The obstacle generates a repulsive force:

- Pushes the robot away from dangerous regions
- Prevents collision

The combined force generates a safe path:



The APF algorithm allows the robot to modify its trajectory automatically when obstacles exist in the workspace.

4. **Simulation Setup**

The simulation environment is designed to represent an industrial pick-and-place application.

The initial configuration includes:

Robot:

Franka Panda 7-DOF Manipulator

Object:

Pick Location: (0.75, -0.25, 0.70)

Target:

Place Location: (0.15, 0.35, 0.70)

Obstacle:

Obstacle Position: (0.50, 0.25, 0.72)

The obstacle is intentionally placed between the starting position and target position to test the effectiveness of the obstacle avoidance algorithm.

5. System Execution Result

The complete simulation process contains the following steps:

➤ Step 1: Move to Object

The robot moves from the home position to the object pickup location.

Result: Successful

➤ Step 2: Grasp Object

The gripper closes and attaches the object.

Result: Object Successfully Attached

➤ Step 3: Lift Object

The robot lifts the object to a safe height before trajectory planning.

Result: Successful

➤ Step 4: Automatic Obstacle Avoidance

The APF algorithm generates a collision-free trajectory around the obstacle.

Result: Collision Avoidance Successful

➤ Step 5: Place Object

The robot slowly lowers the object to the target position and releases it.

Result: Object Successfully Released

➤ Step 6: Return Home

The robot returns to the initial home position.

Result: Task Completed

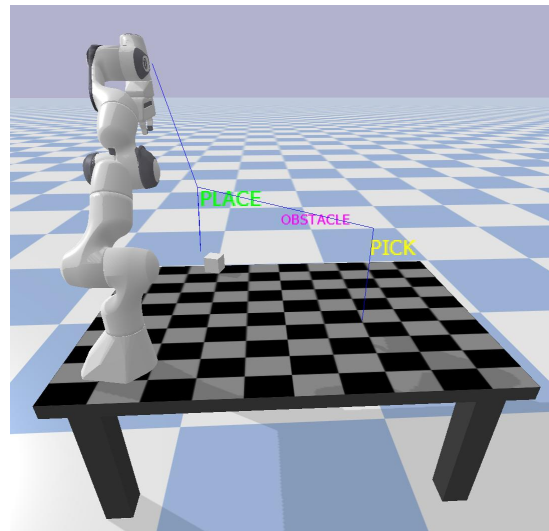
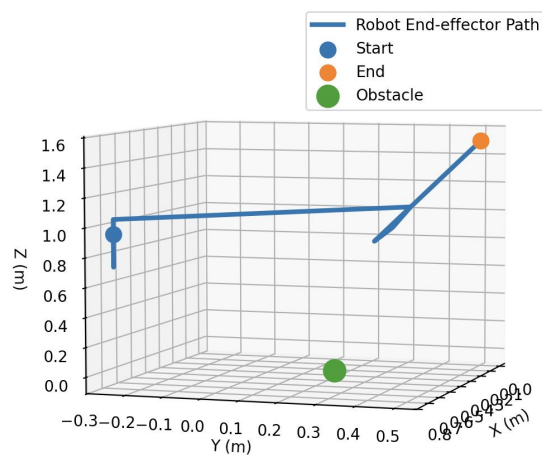
6. Multi-Scenario Performance Evaluation

➤ Scenario 1: Normal Pick-and-Place

Configuration:

Obstacle: None

Robot End-effector Trajectory with APF Obstacle Avoidance



Result:

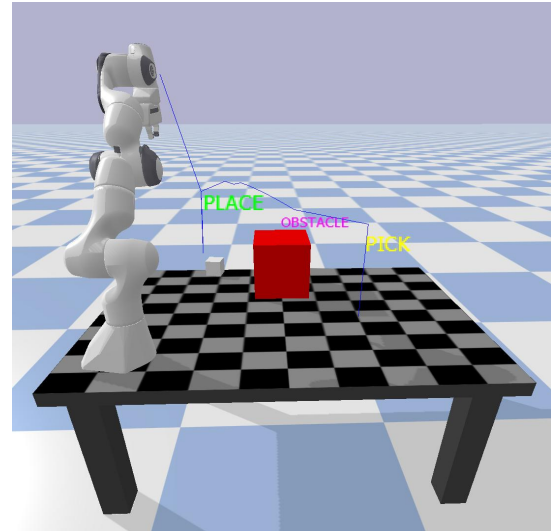
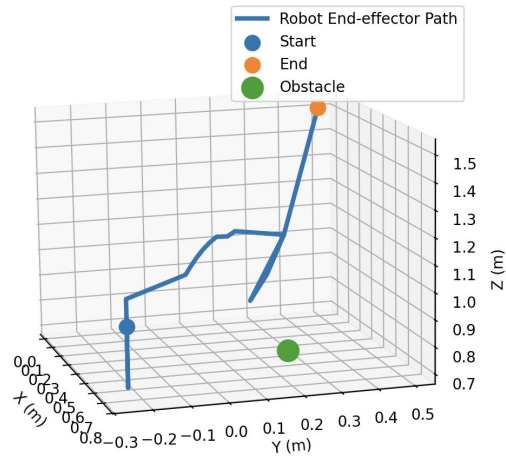
Parameter	Result
Task completion	Successful
Collision	None
Trajectory	Smooth

➤ Scenario 2: Static Obstacle Avoidance

Configuration:

Obstacle Position: (0.50,0.25,0.72)

Robot End-effector Trajectory with APF Obstacle Avoidance



Result:

Parameter	Result
Task completion	Successful
Collision	None
Obstacle avoidance	Successful

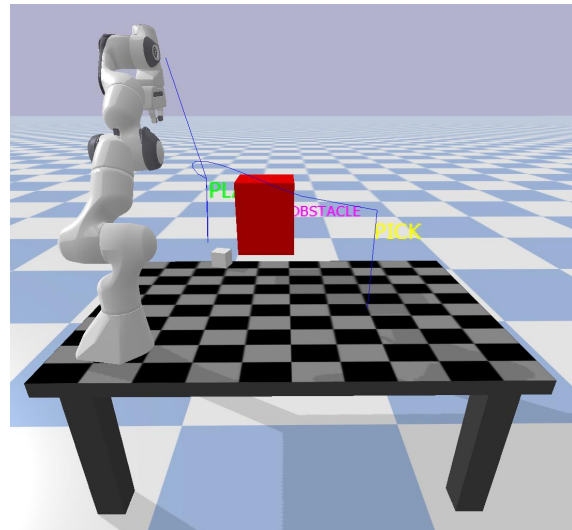
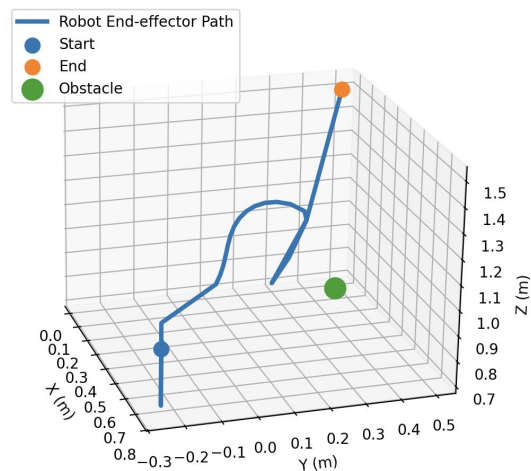
The APF algorithm successfully modified the robot trajectory and avoided collision.

➤ Scenario 3: Different Obstacle Position

Configuration:

Obstacle Position: (0.40,0.35,0.90)

Robot End-effector Trajectory with APF Obstacle Avoidance



Result:

Parameter	Result
Task completion	Successful
Collision	None
Path generation	Successful

The system maintained stable performance with different obstacle configurations.

7. Trajectory Analysis

The generated end-effector trajectory is recorded during simulation.

The trajectory analysis demonstrates:

- Smooth robot movement
- Continuous path generation
- Successful obstacle avoidance

The trajectory plot includes:

- Start position
- Goal position
- Obstacle position
- End-effector movement path

Robot End-effector Trajectory with APF Obstacle Avoidance

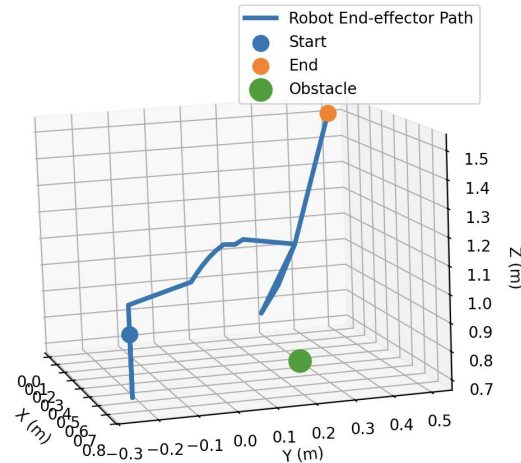


Figure 2. End-effector Trajectory with APF Obstacle Avoidance

8. Discussion

The developed system successfully integrates multiple robotic technologies into a complete manipulation framework.

The main achievements include:

- Integrated robot simulation using PyBullet.
- Cartesian motion planning using trajectory interpolation.
- Robot control using inverse kinematics.
- Automatic obstacle avoidance using APF.
- Fully automated pick-and-place task execution.

The results demonstrate that the robot can perform industrial transportation tasks while maintaining safe and smooth motion.

9. Limitations and Future Improvements

Although the system successfully completes the pick-and-place task, several improvements can be considered.

Current limitations:

- Only static obstacles are considered.
- Object grasping is simplified using fixed constraints.
- No external sensors are used for environment perception.

Future improvements:

- Add camera-based object detection.
- Implement dynamic obstacle avoidance.
- Integrate ROS framework.
- Add force feedback control.
- Deploy algorithms on a real industrial robot.

10. Conclusion

In Stage 4, a complete integrated robot motion planning and pick-and-place simulation system was successfully developed.

The system combines task scheduling, inverse kinematics, trajectory planning, obstacle avoidance, and robot control into a unified framework.

The final PyBullet simulation demonstrates that the robot can autonomously complete an industrial pick-and-place task while avoiding obstacles and maintaining smooth motion.

This project provides a foundation for future development of intelligent industrial robotic systems.